

Preliminaries and Functions

Numbers, Absolute Values, and Inequalities

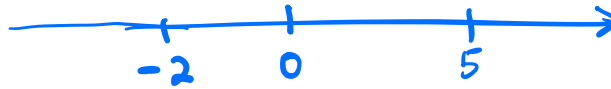
Real Numbers and Real Lines

Much of the calculus is based on properties of the real number system. **Real numbers** are numbers that can be expressed as decimals, such as

$$-\frac{3}{4} = -0.7500 \dots$$

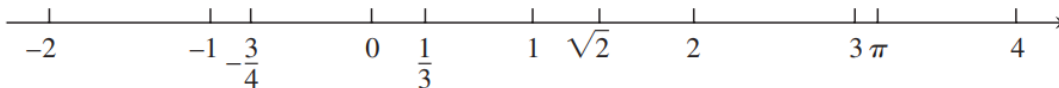
$$\frac{1}{3} = 0.33333 \dots$$

$$\sqrt{2} = 1.4142 \dots$$



Note: The dots “...” in each case indicate the sequence of decimal digits goes forever.

The real numbers can be represented geometrically as points on a number line called **real line**.



We use $\mathbb{R}$ denotes real number system.

$-b \rightarrow$ inverse of b

The properties of the real number system fall into three categories: *Algebraic Properties*, *Order Properties*, and *Completeness*.

$a + (-b)$

- (i) **Algebraic Properties:** the real numbers can be added, subtracted, multiplied, and divided (except by 0), to produce more real numbers under usual rules of arithmetic.

- (ii) **Ordered Properties:**

Rule for Inequality: If a, b and c are real numbers, then

1. $a < b \Rightarrow a + c < b + c$

2. $a < b \Rightarrow a - c < b - c$

3. $a < b$ and $c > 0 \Rightarrow ac < bc$

4. $a < b$ and $c < 0 \Rightarrow bc < ac$

(Special case: $a < b \Rightarrow -b < -a$)

5. $a > 0 \Rightarrow \frac{1}{a} > 0$

6. If a and b are both positive or both negative, then $a < b \Rightarrow \frac{1}{b} < \frac{1}{a}$.

- (iii) **Completeness Property:** Roughly speaking, it says that there are enough real numbers to “complete” the real number line, in the sense that there are no “holes” or “gaps” in it.

Note: The set of rational numbers has all the algebraic and order properties of the real numbers but *lacks the completeness property*.

For example, there is no rational number whose square is 2; there is a “hole” in the rational line where should be. Real numbers that are not rational are called **irrational numbers**.

• function
- limit
- differentiation
- Integration

→ Pre-test - 5%
Tests → 15%
→ 20%
→ 40%
Final - 60%
→ 100%
→ 50%

Intervals

interval

$$\begin{aligned} \text{open } (a,b) &\rightarrow a < x < b \\ \text{closed } [a,b] &\rightarrow a \leq x \leq b \\ (a,b] &\rightarrow a < x \leq b \\ [a,b) &\rightarrow a \leq x < b \end{aligned}$$

A subset of real line is called an **interval** if it contains at least two numbers and contains all the real numbers lying between any two of its elements.

For example, if $a < b$, the open interval from a to b consists of all numbers between a and b and is denoted by the symbol (a,b) . Using set builder notation, we write

$$(a,b) = \{x \mid a < x < b\}.$$

Notice that the endpoints of the interval, namely, a and b are excluded.

The *closed interval* from a to b consists of all numbers between a and b , including a and b . We denote this closed interval by $[a,b]$. Using set builder notation, we write

$$[a,b] = \{x \mid a \leq x \leq b\}.$$

$$x \geq a \quad [a, \infty)$$

$$x < a \quad (-\infty, a)$$

Intervals can also be half-open (or half-closed), e.g.

$$[a,b] = \{x \mid a < x \leq b\}.$$

$$(-\infty, \infty) \rightarrow x \in \mathbb{R}$$

We also consider infinite intervals such as

$$[a, \infty) = \{x \mid x \geq a\}, \quad (-\infty, b) = \{x \mid x < b\}, \quad (-\infty, \infty) = \{x \mid x \in \mathbb{R}\}.$$

$\mathbb{R}$
real number

The symbol ∞ or $-\infty$ are not numbers. For example, ∞ simply indicates that the interval extends indefinitely far in the positive direction.

Notation	Set description	Picture
(a,b)	$\{x \mid a < x < b\}$	
$[a,b]$	$\{x \mid a \leq x \leq b\}$	
$[a,b)$	$\{x \mid a \leq x < b\}$	
$(a,b]$	$\{x \mid a < x \leq b\}$	
(a, ∞)	$\{x \mid x > a\}$	
$[a, \infty)$	$\{x \mid x \geq a\}$	
$(-\infty, b)$	$\{x \mid x < b\}$	
$(-\infty, b]$	$\{x \mid x \leq b\}$	
$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	

Table of Intervals

Example. Solve the inequality $\frac{x+1}{x-1} < 2$.

Solution:

$$\frac{x+1}{x-1} < 2 \quad \begin{aligned} &\rightarrow x-1 > 0 \\ &\rightarrow x-1 < 0 \end{aligned}$$

$$\begin{aligned} \frac{x+1}{x-1} - 2 &< 0 \\ \frac{x+1-2(x-1)}{x-1} &< 0 \end{aligned}$$

$$\frac{-x+3}{x-1} < 0$$

$$\frac{3-x}{x-1} < 0$$

$$x < 1, \quad x > 3.$$

Absolute Value

The *absolute value* of a number a , denoted $|a|$, is the distance from a to 0 on the real line, given a convenient unit of measurement. Distances are always non-negative, so we have

$$|a| \geq 0 \text{ for any number } a.$$

To facilitate algebraic manipulation, we write the absolute value function as a step function:

$$\sqrt{a^2} = |a| = \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

Let a and b be real numbers, and n be an integer. We have the following properties of absolute values:

1. $|a| = \sqrt{a^2}$. $\rightarrow \sqrt{a^2} = (\sqrt{a})^2 \quad a \geq 0$
2. $|ab| = |a||b|$.
3. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$, $b \neq 0$.
4. $|a^n| = |a|^n$.
5. Suppose that $a > 0$. Then $|x| < a \Leftrightarrow -a < x < a$.
6. Suppose that $a > 0$. Then $|x| > a \Leftrightarrow x < -a \text{ or } x > a$.

The Triangle Inequality. If a and b be real numbers, then $|a + b| \leq |a| + |b|$.

Example. Solve the inequality $|3x - 2| \leq |x - 1|$.

Solution

$$\sqrt{(3x-2)^2} \leq \sqrt{(x-1)^2}$$

$$(3x-2)^2 \leq (x-1)^2$$

$$9x^2 - 12x + 4 \leq x^2 - 2x + 1$$

$$8x^2 - 10x + 3 \leq 0$$

$$(4x-3)(2x-1) \leq 0$$

$$(4x-3)(2x-1) \leq 0$$

Let $4x-3 \geq 0$

$$x \geq \frac{3}{4}$$

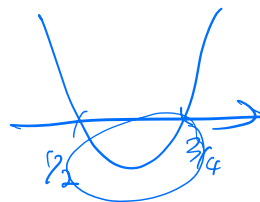
$$2x-1 \geq 0$$

$$x \geq \frac{1}{2} \rightarrow$$

$$\frac{1}{2} \leq x \leq \frac{3}{4}$$

$$|x-1| = \begin{cases} x-1, & x \geq 1 \\ -(x-1), & x < 1 \end{cases}$$

$$|3x-2| = \begin{cases} 3x-2, & x \geq \frac{2}{3} \\ -(3x-2), & x < \frac{2}{3} \end{cases}$$



Example. Describe the set of points satisfying the inequality $x^2 + y^2 - 4x - 6y < 12$.

Solution

$$(x^2 - 4x) + (y^2 - 6y) < 12.$$

$$(x-2)^2 - 4 + (y-3)^2 - 9 < 12$$

$$(x-2)^2 + (y-3)^2 < 25 \equiv 5^2.$$

Disk.



Extra Examples:

1. Solve the inequality $x^2 - 5x + 6 \leq 0$.

2. Solve $x^3 + 3x^2 > 4x$.

3. Solve the inequality.

a. $|x + 3| = |2x + 1|$

b. $|x + 1| \geq 3$

c. $1 \leq |x| \leq 4$

4. Prove that $|ab| = |a||b|$.

Tues 10-11 am, 11-12 pm BTM4

Thurs 10-11 am, 11-12 pm BTM4

Quiz - 5% w4

Test 1 - 15% w7 (online).

Test 2 - 20% w11/w12

select 1 group.

Functions and Their Graphs



Functions; Domain and Range (pg 21)

The value of one variable quantity, say y , depends on the value of another variable quantity, which we often call x . Since the value of y is completely determined by the value x , we say that y is a function of x .

$$A = \pi r^2$$

Often the value of y is given by a *rule* or formula that says how to calculate it from the variable x .

$$A = f(r)$$

Example: The equation $A = \pi r^2$ is a rule that calculates the area A of a circle from its radius r .

$$y = f(x)$$

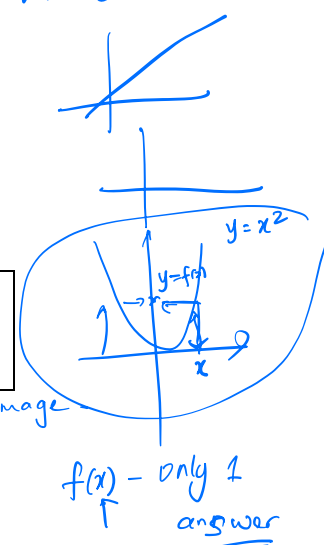
In calculus, we may want to refer to an unspecified function without having any particular formula in mind.

↑ dependent variable.
↑ independent variable.

A symbolic way to say “ y is a function of x ” is by writing

$$y = f(x)$$

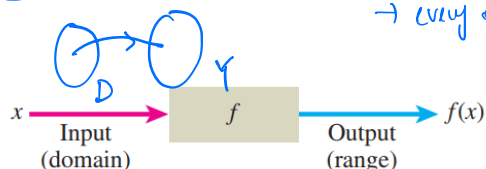
In this notation, the symbol f represents the function. The letter x , called the **independent variable**, represents the input value of f , and y , the **dependent variable**, represents the corresponding output value of f at x .



Definition:

A **function** f from a set D to a set Y is a rule that assigns a single (unique) element $f(x)$ in Y to each $x \in D$.

→ every element in domain, it has 1 and only 1 image



The set D of all possible input values is called the **domain** of the function.

The set of all output values of $f(x)$ as x varies throughout D is called the **range** of the

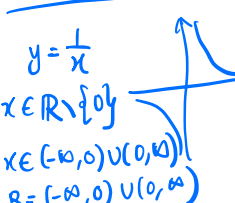
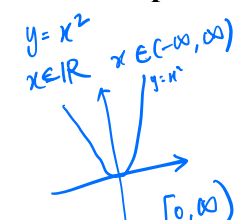
function.

The range may not include every element in the set Y .

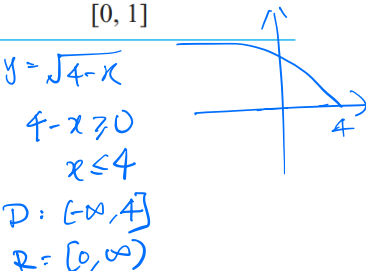
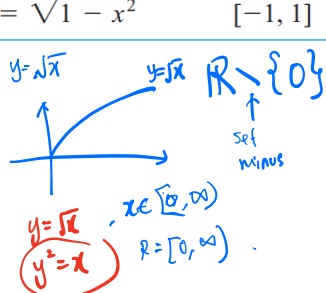
The domain and range of a function can be any sets of objects, but often in calculus they are sets of real numbers interpreted as points of a coordinate line.

∈ element of

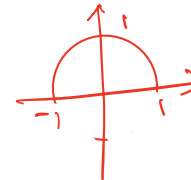
Example. Verify the domains and ranges of these functions.



Function	Domain (x)	Range (y)
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = 1/x$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4-x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1-x^2}$	$[-1, 1]$	$[0, 1]$



$$y = \sqrt{1-x^2} \Rightarrow y^2 = 1-x^2 \Rightarrow x^2 + y^2 = 1$$



$$D = [-1, 1] \\ R = [0, 1] \\ 1-x^2 \geq 0 \Rightarrow x^2 \leq 1 \Rightarrow |x| \leq 1 \Rightarrow -1 \leq x \leq 1 \\ \sqrt{x^2} \leq 1$$

Example (Exercises 1.1).

Find the domain of $f(x) = 1 - \sqrt{x}$, $g(x) = \sqrt{x^2 - 3x}$ and $G(t) = \frac{2}{t^2 - 16}$

Solution

$$f(x) = 1 - \sqrt{x}$$

$$D = x \in [0, \infty)$$

$$g(x) = \sqrt{x^2 - 3x} \quad D = (-\infty, 0] \cup [3, \infty)$$

$$x^2 - 3x \geq 0$$

$$x(x-3) \geq 0$$

Let $x \geq 0$ $x-3 \geq 0$

$$x \geq 0 \quad x \geq 3$$

$$= \begin{array}{c|c|c} - & + & + \\ \hline + & 0 & 3 & + \end{array}$$

$$x \leq 0, x \geq 3$$

$$G(t) = \frac{2}{t^2 - 16}$$

$$= \frac{2}{(t-4)(t+4)}$$

$$t \neq 4, -4$$

$$D = \mathbb{R} \setminus \{-4, 4\}$$

$$(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$$

Graphs of Function (pg 23)

If f is a function with domain D , its **graph** consists of the points in the *Cartesian plane* whose coordinates are the input-output pairs for f . In set notation, the graph is

$$\{(x, f(x)) \mid x \in D\}.$$

input output

The graph of a function $f(x) = x + 2$ is the set of points with coordinates (x, y) for which $y = x + 2$.

The graph of a function f is a useful picture of its behavior. If (x, y) is a point on the graph, then $y = f(x)$ is the height of the graph above the point x . The height may be positive or negative, depending on the sign of $f(x)$.

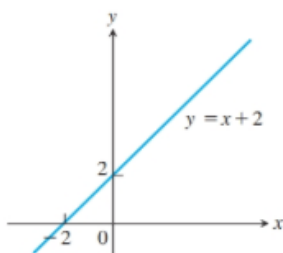


FIGURE 1.3 The graph of $f(x) = x + 2$ is the set of points (x, y) for which y has the value $x + 2$.

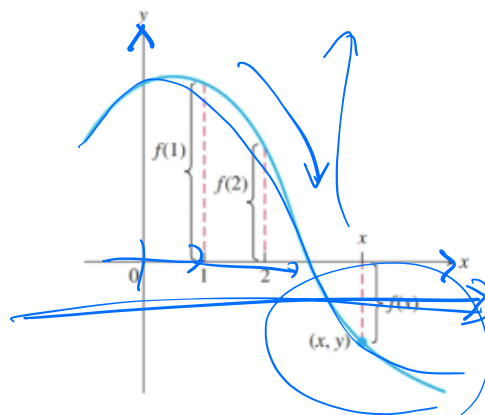
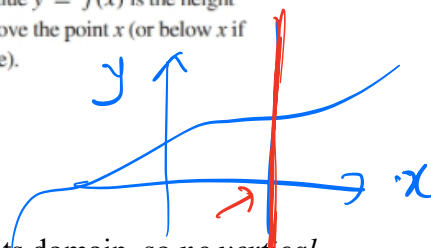


FIGURE 1.4 If (x, y) lies on the graph of f , then the value $y = f(x)$ is the height of the graph above the point x (or below x if $f(x)$ is negative).

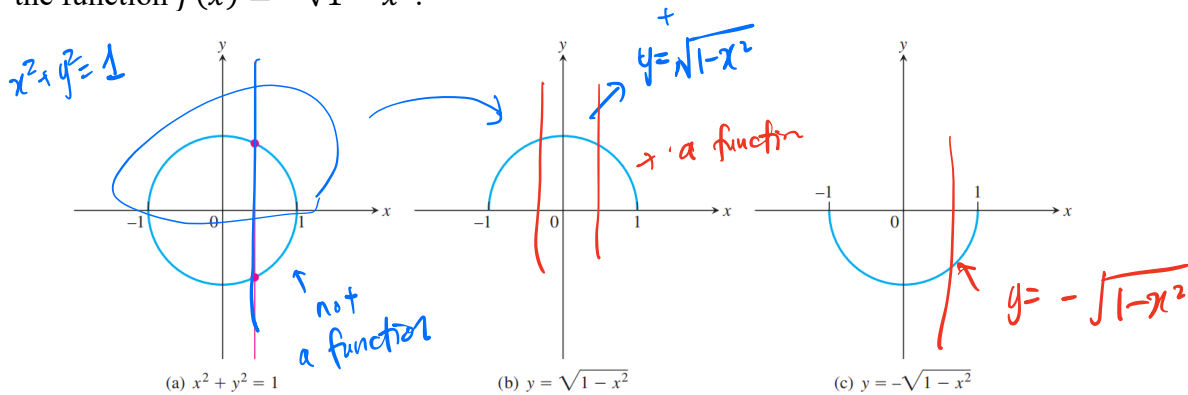
The Vertical Line Test for a Function (pg 24)

A function f can have only **one value** $f(x)$ for each x in its domain, so *no vertical line* can intersect the graph of a function more than one point. If a is in the domain of a function f , then the vertical line $x = a$ will intersect the graph of f in the single point $(a, f(a))$.



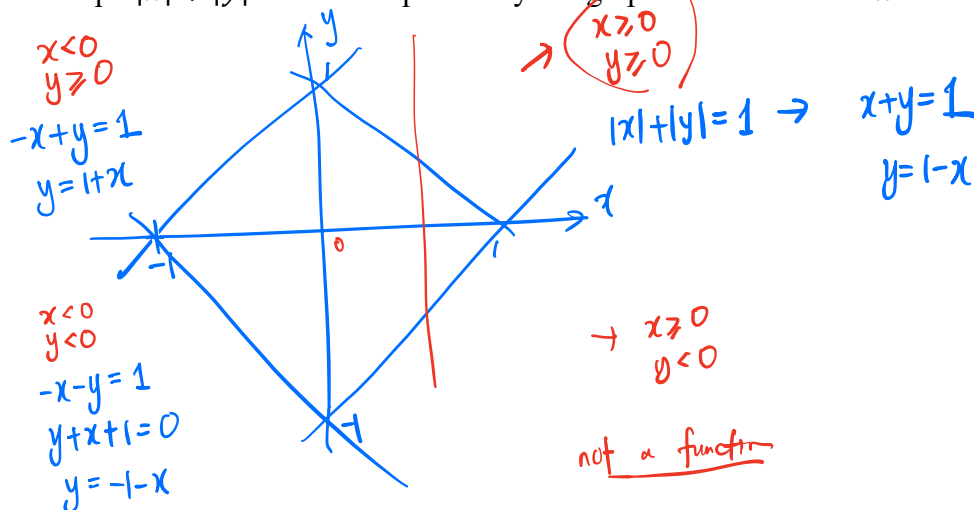
$$x^2 + y^2 = 1$$

Example. The circle cannot be the graph of a function, since some vertical lines intersect the circle twice. The circle does contain the graphs of two functions of x ; namely the upper semicircle defined by the function $f(x) = \sqrt{1 - x^2}$ and the lower semicircle defined by the function $f(x) = -\sqrt{1 - x^2}$.



Examples (Exercises 1.1):

Graph $|x| + |y| = 1$ and explain why not graphs of functions of x .



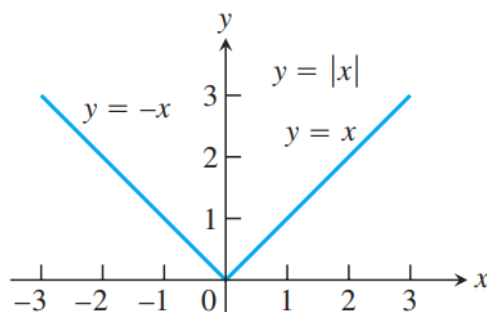
Piecewise-Defined Functions

Sometimes a function is described by using different formulas on different parts of its domain.

Example. Absolute Value Function.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0, \end{cases}$$

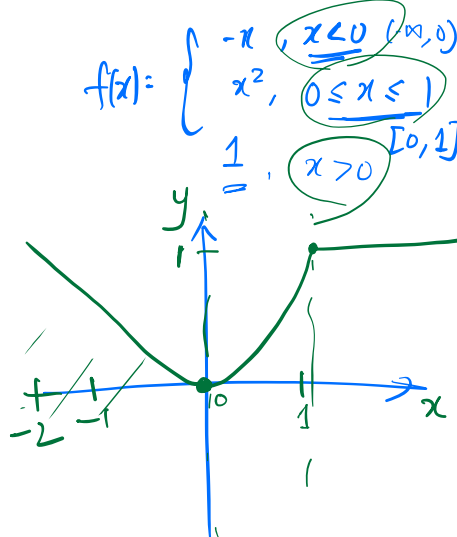
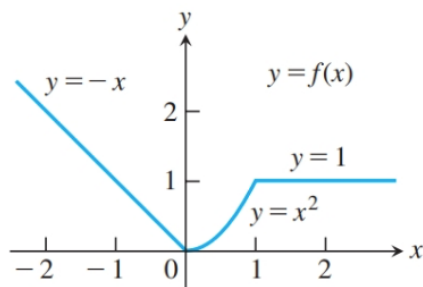
$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



The absolute value function has domain $(-\infty, \infty)$ and range $[0, \infty)$.

Example. The function

$$f(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

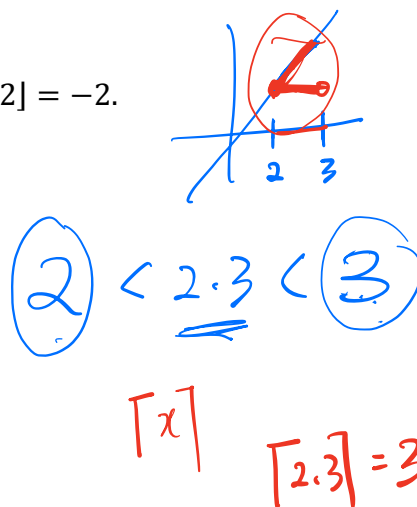
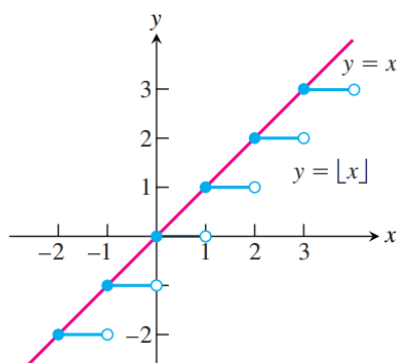


To graph the function $y = f(x)$, we apply different formulas to different parts of its domain.

Example. The Greatest Integer Function (Integer Floor Function)

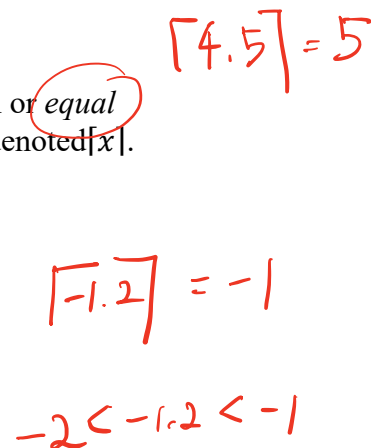
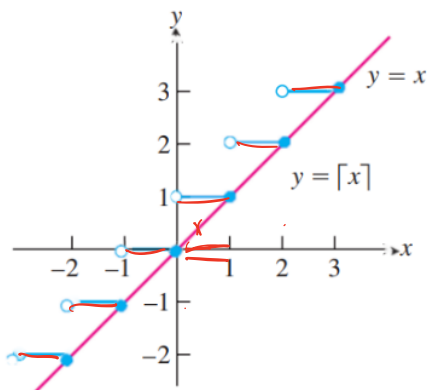
The function whose value at any number x is the greatest integer less than or equal to x is called the **greatest integer function** or the **integer floor function**. It is denoted $\lfloor x \rfloor$. Observe that

$$\lfloor 2.4 \rfloor = 2, \lfloor 1.9 \rfloor = 1, \lfloor 0 \rfloor = 0, \lfloor 2 \rfloor = 2, \lfloor -1.2 \rfloor = -2, \lfloor -0.3 \rfloor = -1, \lfloor -2 \rfloor = -2.$$



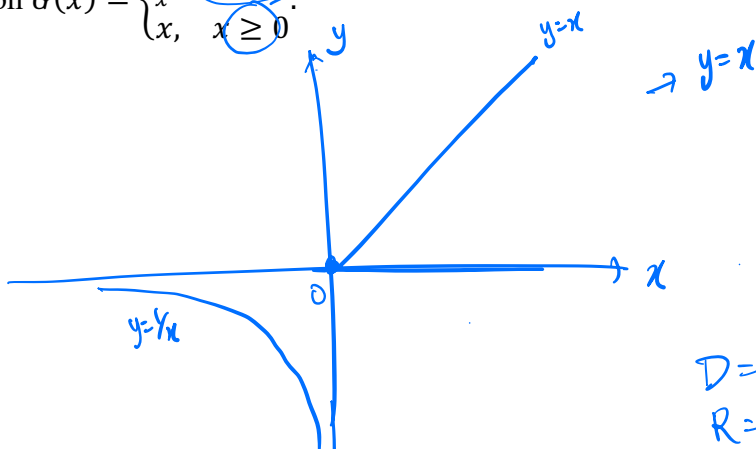
Example. The Least Integer Function (Integer Ceiling Function)

The function whose value at any number x is the smallest integer greater than or equal to x is called the **least integer function** or the **integer ceiling function**. It is denoted $\lceil x \rceil$.



Examples:

Graph $G(x) = \begin{cases} \frac{1}{x}, & x < 0 \\ x, & x \geq 0 \end{cases}$



Increasing and Decreasing functions

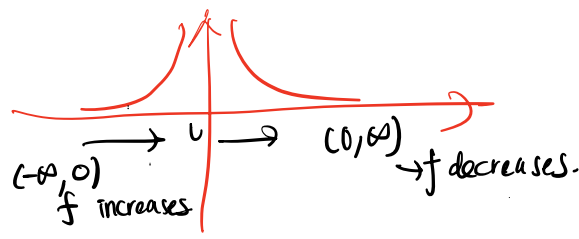
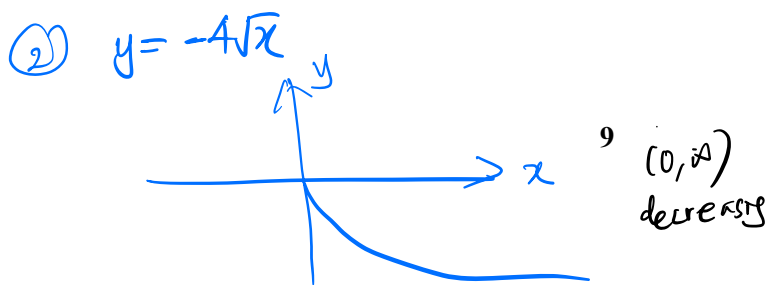
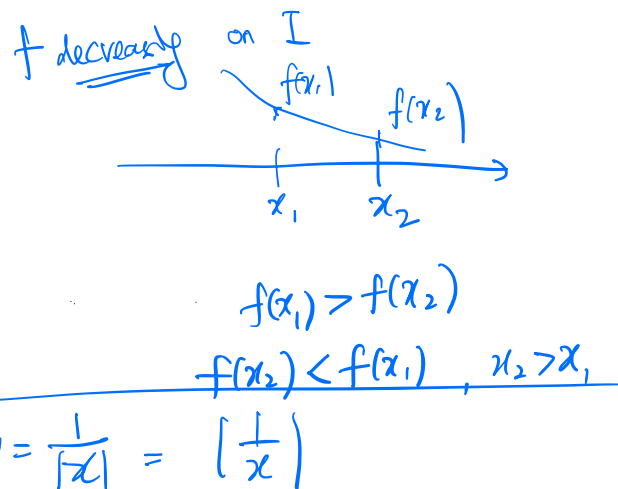
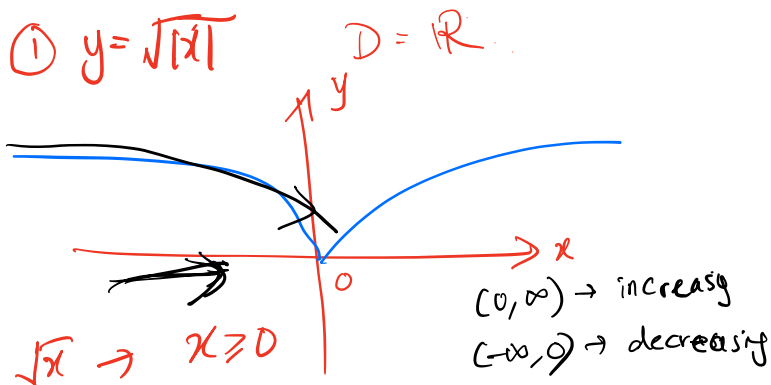
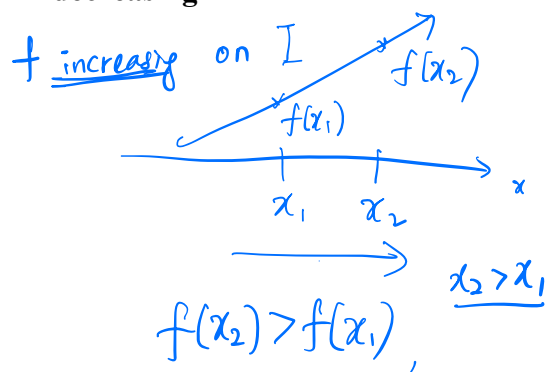
If the graph of a function climbs or rises as you move from left to right, we say that the function is *increasing*. If the graph descends or falls as you move from left to right, the function is *decreasing*.

Definitions Let f be a function defined on an interval I and let x_1 and x_2 be two distinct points in I .

1. If $f(x_2) > f(x_1)$ whenever $x_1 < x_2$, then f is said to be **increasing** on I .
2. If $f(x_2) < f(x_1)$ whenever $x_1 < x_2$, then f is said to be **decreasing** on I .

Examples:

1. $y = \sqrt{|x|}$
2. $y = -4\sqrt{x}$
3. $y = \frac{1}{|x|}$



$$f(x) = x^2$$

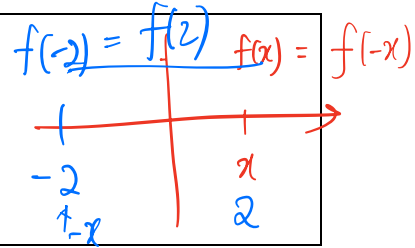
Definitions. Even functions and Odd functions: Symmetry

A function $y = f(x)$ is an

even function of x if $f(-x) = f(x)$,

odd function of x if $f(-x) = -f(x)$,

for every x in the function's domain.

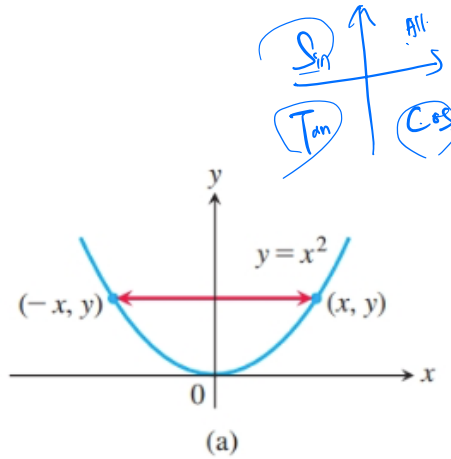
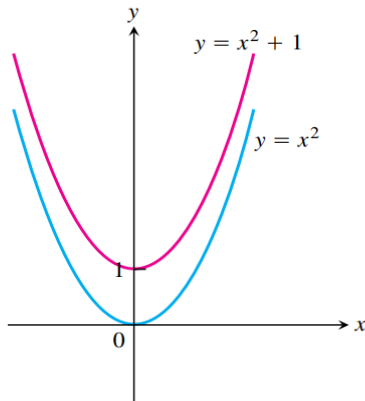


The graph of an *even function* is **symmetric about the y-axis** ($f(-x) = f(x)$).

Note:

1. a point (x, y) lies on the graph if and only if the point $(-x, y)$ lies on the graph.
2. a reflection across the y-axis leaves the graph unchanged.

Example: $y = x^2$, $y = x^2 + 1$.

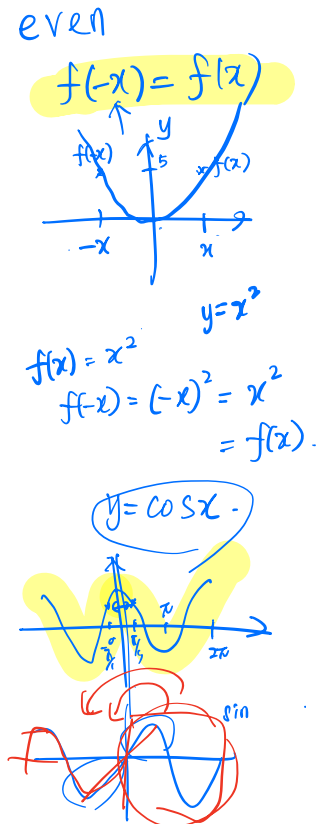
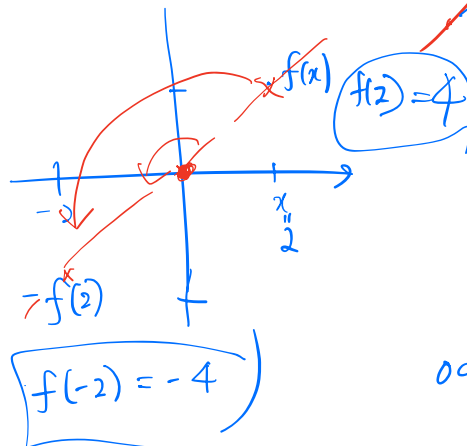
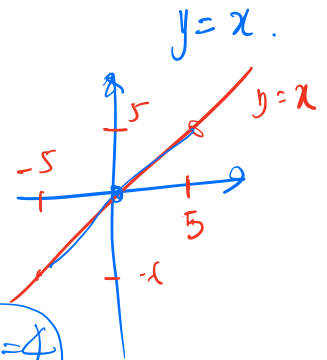
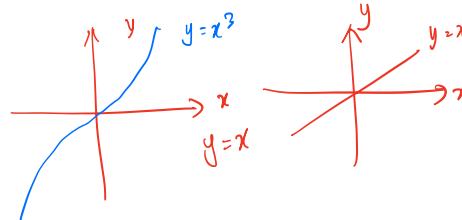
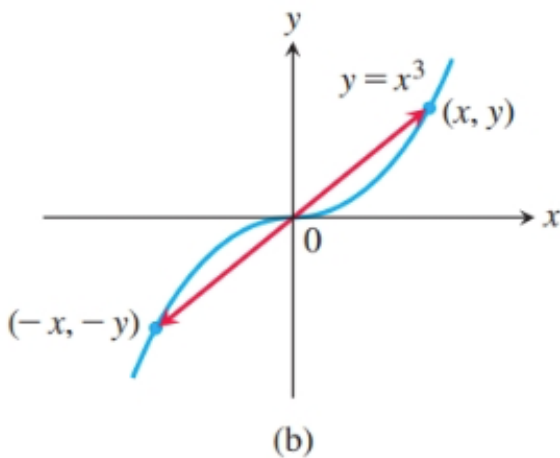


The graph of an *odd function* is **symmetric about the origin** ($f(-x) = -f(x)$).

Note:

1. a point (x, y) lies on the graph if and only if the point $(-x, -y)$ lies on the graph.
2. a graph is symmetric about the origin if a rotation of 180° about the origin leaves the graph unchanged.
3. Notice that the definitions imply both x and $-x$ must be in the domain of f .

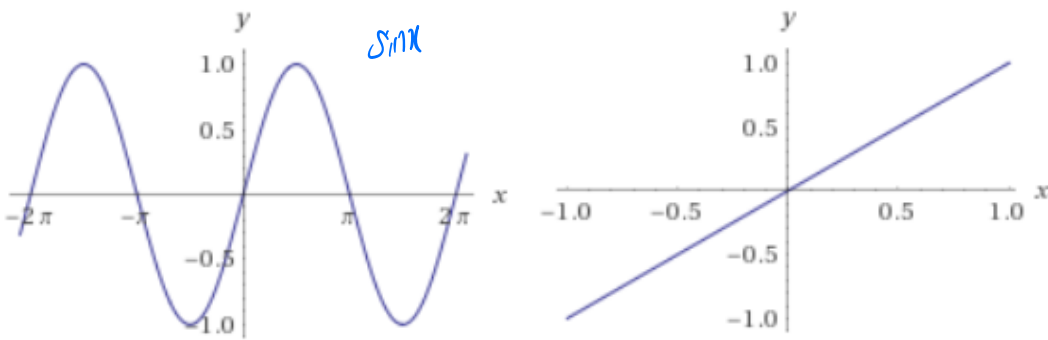
Example: $y = x^3$, $y = x$, $y = \sin x$.



odd
 $f(-x) = -f(x)$

$$f(-2) = -f(2)$$

odd.



Examples:

1. $y = x^2 + x$

2. $y = 2|t| + 1$

3. $y = \frac{x}{x^2 - 1}$

4. $y = \sin 2x$

even $f(-x) = f(x)$, odd $f(-x) = -f(x)$

$f(x) = x^2 + x$, $f(-x) = (-x)^2 + (-x) = x^2 - x \neq f(x)$
not even not odd.

$f(t) = 2|t| + 1$, $f(-t) = 2|-t| + 1 = 2|t| + 1 = f(t)$
even

$f(x) = \frac{x}{x^2 - 1}$

$f(-x) = \frac{f(x)}{(-x)^2 - 1} = \frac{-x}{x^2 - 1} = -\left(\frac{x}{x^2 - 1}\right)$

$= -f(x)$
odd.

$f(x) = \sin 2x$

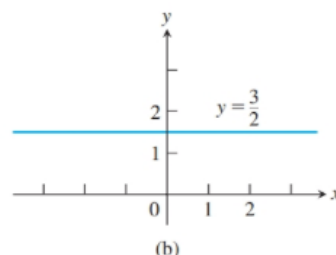
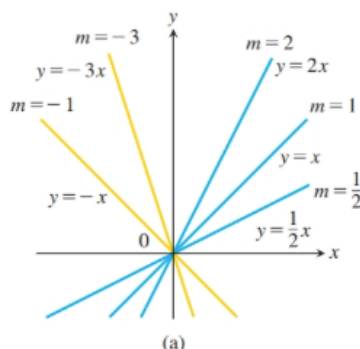
$f(-x) = \sin(-2x) = -\sin 2x = -f(x)$
odd.

Common Functions.

Linear Functions. $y = mx + b$ for constants m and b .

$f(x) = x$

identity function



$0x + 3/2$

FIGURE 1.14 (a) Lines through the origin with slope m . (b) A constant function with slope $m = 0$.

Power Functions. $f(x) = x^a$, where a is a constant.

(a) $a = n$, a positive integer.

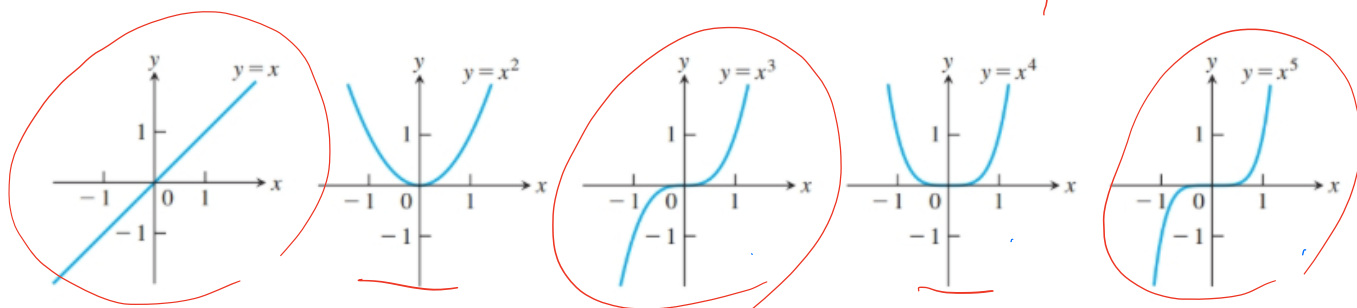
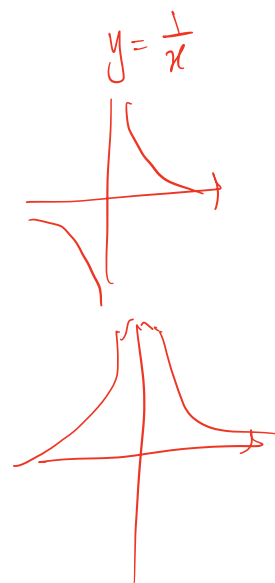
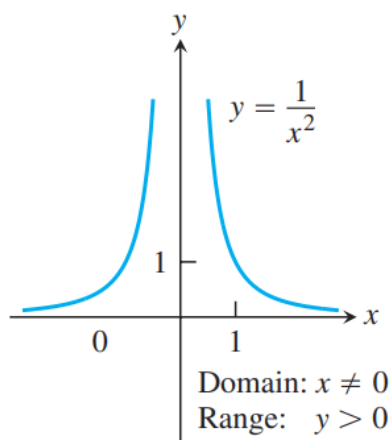
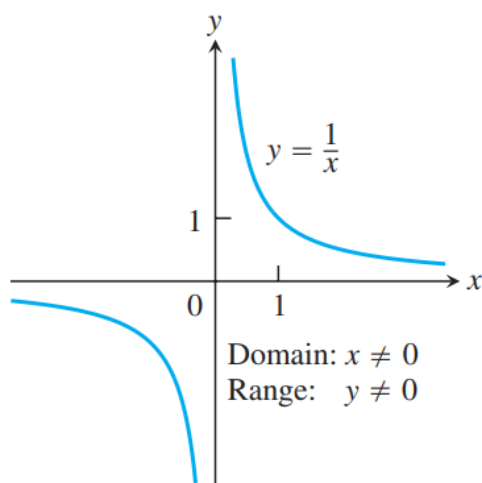


FIGURE 1.15 Graphs of $f(x) = x^n$, $n = 1, 2, 3, 4, 5$, defined for $-\infty < x < \infty$.

(b) $a = -1$, $a = -2$.

The graph of the functions $f(x) = x^{-1} = \frac{1}{x}$ and $g(x) = x^{-2} = \frac{1}{x^2}$, where $x \neq 0$.



(c) $a = \frac{1}{2}, \frac{1}{3}, \frac{3}{2}, \frac{2}{3}$

$x^{1/2}, x^{1/3}, x^{3/2}, x^{2/3}$

We note $f(x) = x^{1/2} = \sqrt{x}$ (square root) and $g(x) = x^{1/3} = \sqrt[3]{x}$ (cube root) with domain $[0, \infty)$, but $g(x) = x^{1/3} = \sqrt[3]{x}$ is defined for all real x . Recall that $x^{3/2} = (x^{1/2})^3$ and $x^{2/3} = (x^{1/3})^2$.

$y = x^{1/3}$
 $y = \sqrt{x}$
 $y^2 = x$

$x^{3/2} = (x^{1/2})^3$
 $x^{2/3} = (x^{1/3})^2$

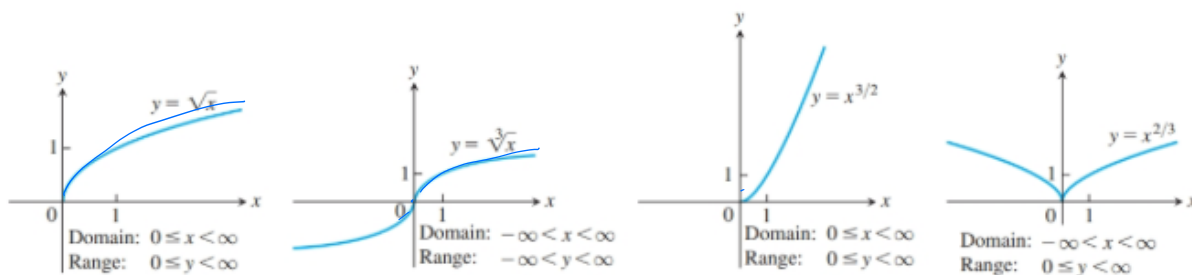


FIGURE 1.17 Graphs of the power functions $f(x) = x^a$ for $a = \frac{1}{2}, \frac{1}{3}, \frac{3}{2}$, and $\frac{2}{3}$.



Polynomials A function $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where n is a non-negative integer and the numbers $a_0, a_1, a_2, \dots, a_n$ are real constants (**coefficients** of the polynomial).

$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$
power series.

Polynomial
 $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

$$f(x) = 3x^4 + 2x + 1$$

leading coefficient = 3
deg $f(x) = 4$

a_n = leading coefficient, $a_n \neq 0$.
 n = degree of polynomial.

Note:

1. All polynomials have domain $(-\infty, \infty)$.
2. If the leading coefficient $a_n \neq 0$ and $n > 0$, then n is called the **degree** of the polynomial.

$$\sin x \approx x$$

$$\cos x \approx 1 - \frac{x^2}{2}$$

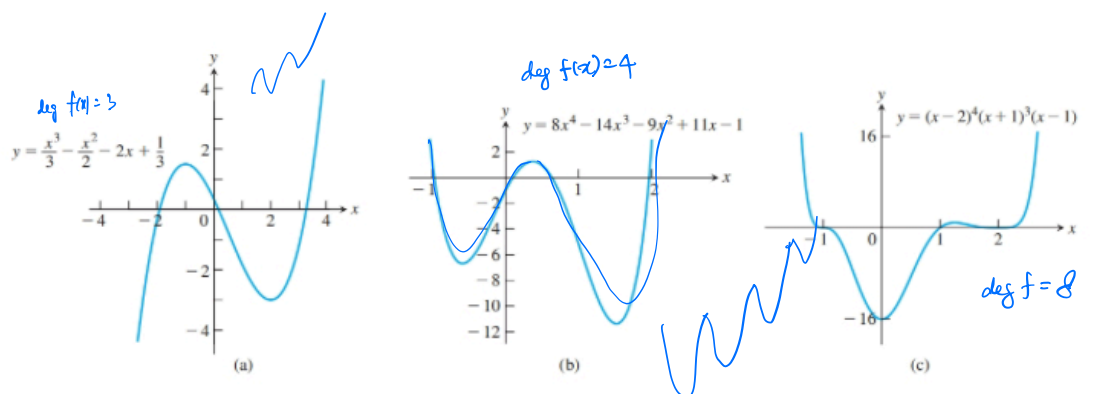
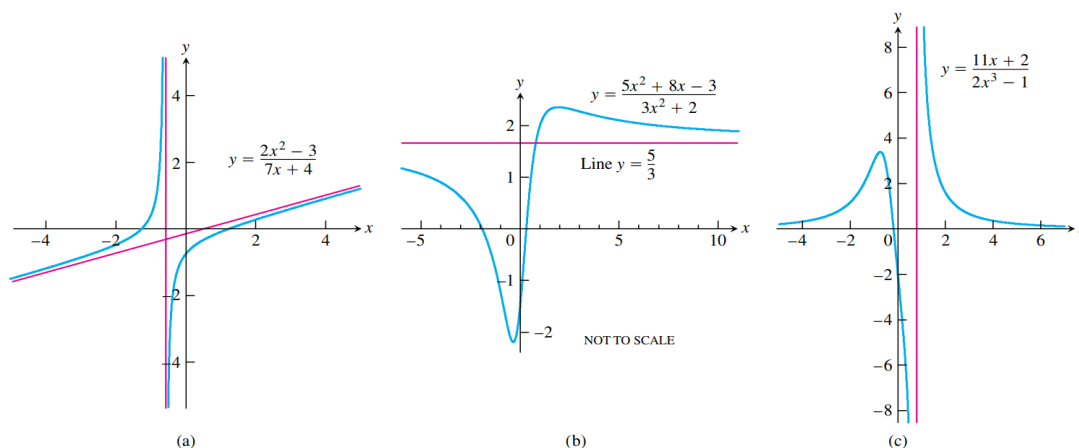


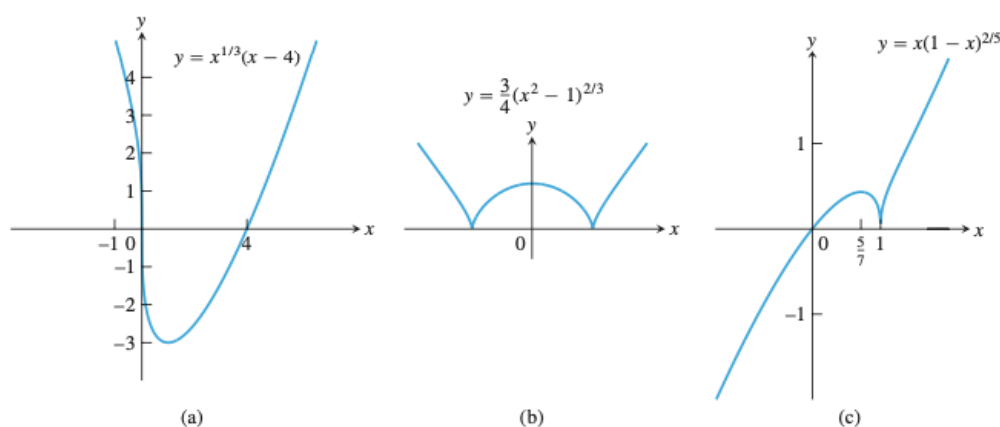
FIGURE 1.18 Graphs of three polynomial functions.

Rational Function. $f(x) = \frac{p(x)}{q(x)}$, where p, q are polynomials. The domain of a rational function is the set of all real x for which $q(x) \neq 0$.

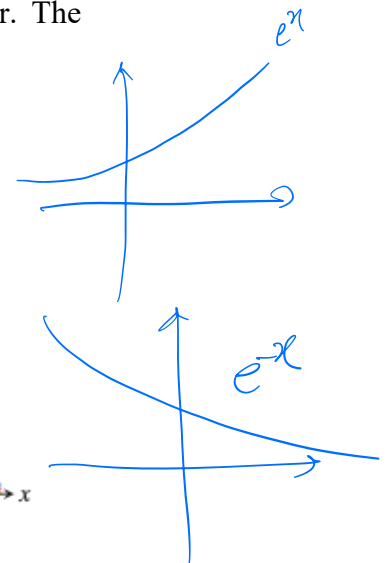
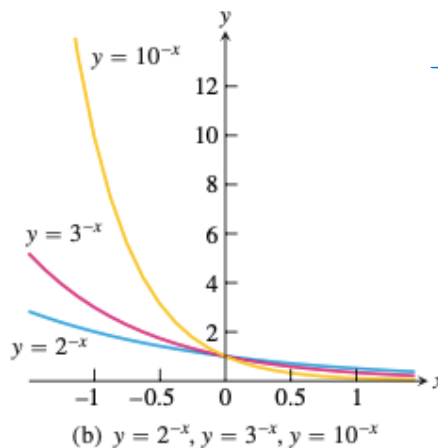
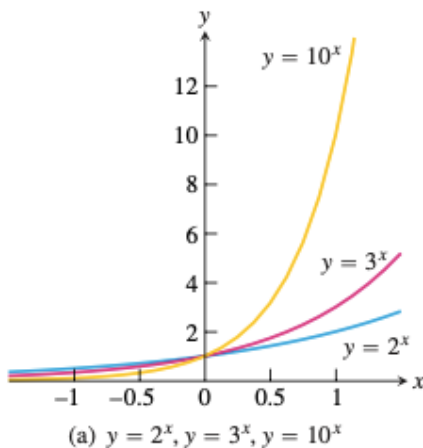
Example. $f(x) = \frac{2x^2-3}{7x+4}$, $x \neq -\frac{4}{7}$. Domain of f is $\{x \mid x \neq -\frac{4}{7}\}$.



Algebraic functions Any function constructed from polynomials using the operations of algebra (addition, subtraction, multiplication, division, and roots) on polynomials or rational functions.

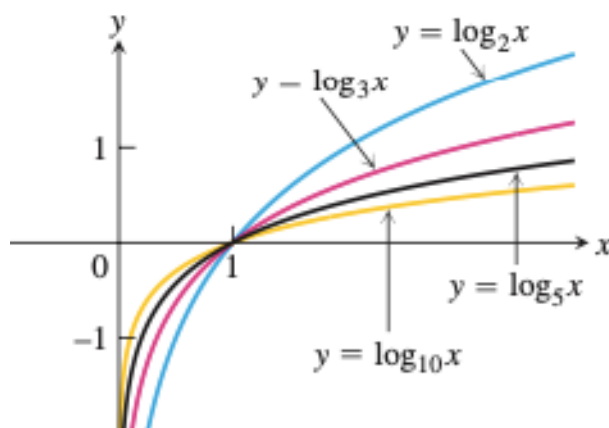


Exponential functions $f(x) = a^x$, where base $a \neq 1$ is a positive real number. The domain is $\mathbf{R}$ and range $(0, \infty)$.
 Note: Exponential function never assumes the value 0.



Logarithmic functions $f(x) = \log_a x$, where the base $a > 0, a \neq 1$. The domain is $(0, \infty)$.

Note: They are inverse functions of the exponential functions.

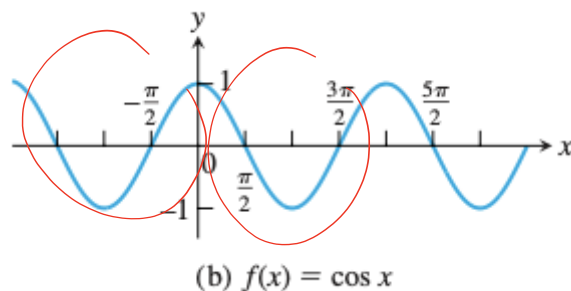
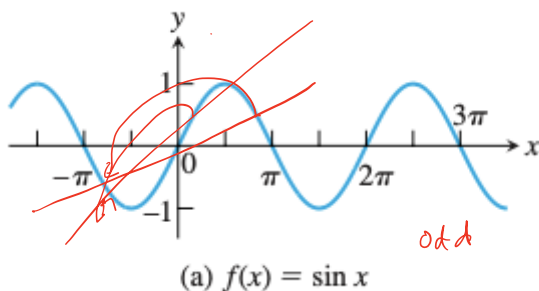


natural log.
 $\ln x$ ~~$\log x$~~

Trigonometric functions

Example. $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$.

radius



Transcendental Functions These are functions that are not algebraic. They include the trigonometric, inverse trigonometric, exponential, and logarithmic functions, and many other functions as well.

Examples:

- Does $\lceil -x \rceil = -\lfloor x \rfloor$ for all real x ?

$$\underline{n < x < n+1}, \quad n \in \mathbb{Z}$$

$$\lfloor x \rfloor = n$$

$$-x \rightarrow -(n+1) < -x < -n$$

$$\lceil -x \rceil = -n$$

$$\lceil -x \rceil = -\lfloor x \rfloor \Rightarrow \text{true for all real } x.$$

Combining Functions; Shifting and Scaling Graphs (pg 14)

Sums, Differences, Products, and Quotients

If f and g are functions, then for every x that belongs to the domains of both f and g (that is, for $x \in D(f) \cap D(g)$), we define functions $f + g$, $f - g$ and fg by the formulas

$$\left. \begin{aligned} (f+g)(x) &= f(x) + g(x) \\ (f-g)(x) &= f(x) - g(x) \\ (fg)(x) &= f(x)g(x). \end{aligned} \right\} D = D_f \cap D_g$$

At any point of $D(f) \cap D(g)$ at which $g(x) \neq 0$, we can also define the function f/g by the formula

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \text{ (where } g(x) \neq 0\text{)}.$$

$$D_{f/g} = D_f \cap D_g, \quad \underline{g(x) \neq 0}.$$

Functions can also be multiplied by constants: If c is a real number, then the function cf is defined for all x in the domain of f by

$$(cf)(x) = cf(x). \quad D_f$$

Example: The functions defined by the formulas

$f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$ have domains $D(f) = [0, \infty)$ and $D(g) = (-\infty, 1]$.

$$D_f = [0, \infty)$$

$$\begin{aligned} 1-x &\geq 0 \\ x &\leq 1 \\ (-\infty, 1] \end{aligned}$$

$$\frac{(fg)(x)}{(f/g)(x)} \rightarrow f \circ g$$

15

$$(fg)(x) = \sqrt{x} \cdot \sqrt{1-x} = \sqrt{x(1-x)} = \sqrt{x-x^2}$$

$$\boxed{\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}}$$

$$(f+g)(x) = \sqrt{x} + \sqrt{1-x}$$

$$D_{f+g} = D_f \cap D_g = [0, \infty) \cap (-\infty, 1] = [0, 1]$$

Function	Formula	Domain
$f + g$	$(f + g)(x) = \sqrt{x} + \sqrt{1-x}$	$[0, 1] = D(f) \cap D(g)$
$f - g$	$(f - g)(x) = \sqrt{x} - \sqrt{1-x}$	$[0, 1]$
$g - f$	$(g - f)(x) = \sqrt{1-x} - \sqrt{x}$	$[0, 1]$
$f \cdot g$	$(f \cdot g)(x) = f(x)g(x) = \sqrt{x(1-x)}$	$[0, 1]$
$\frac{f}{g}$	$\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \sqrt{\frac{x}{1-x}}$	$[0, 1)$ ($x = 1$ excluded)
$\frac{g}{f}$	$\frac{g}{f}(x) = \frac{g(x)}{f(x)} = \sqrt{\frac{1-x}{x}}$	$(0, 1]$ ($x = 0$ excluded)

$$\frac{ab+c}{a} = \frac{ab}{a} + \frac{c}{a}$$

$$\frac{ab+ac}{a} = b+c$$

$$\frac{\sqrt{x}}{\sqrt{1-x}} = \sqrt{\frac{x}{1-x}}$$

$$D_{f/g} = [0, 1)$$

$$\frac{g}{f} = \sqrt{\frac{1-x}{x}} \quad D_{g/f} = (0, 1]$$

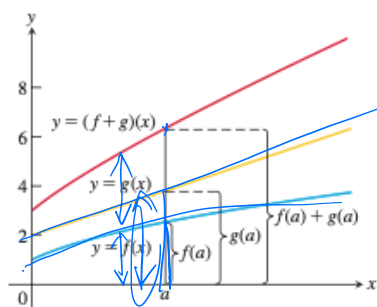


FIGURE 1.25 Graphical addition of two functions.

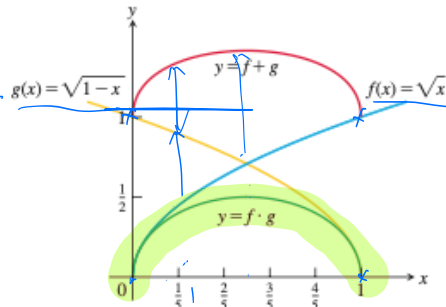
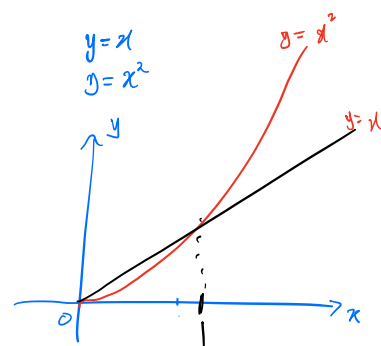


FIGURE 1.26 The domain of the function $f + g$ is the intersection of the domains of f and g , the interval $[0, 1]$ on the x -axis where these domains overlap. This interval is also the domain of the function $f \cdot g$ (Example 1).



Composing Functions

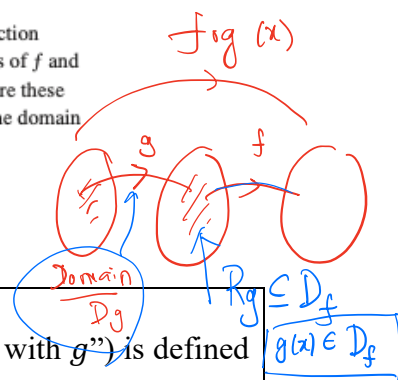
$$f \circ g(x) = f(g(x))$$

Definition.

If f and g are functions, the composite function $f \circ g$ ("f composed with g") is defined by

$$(f \circ g)(x) = f(g(x)).$$

The domain of $f \circ g$ consists of the numbers x in the domain of g for which $g(x)$ lies in the domain of f .



Note:

1. $f \circ g$ can be formed when the range of g lies in the domain of f .
2. Find $(f \circ g)(x)$, first find $g(x)$ and second find $f(g(x))$.



$$x \rightarrow g \rightarrow g(x) \rightarrow f \rightarrow f(g(x))$$

FIGURE 1.27 A composite function $f \circ g$ uses the output $g(x)$ of the first function g as the input for the second function f .

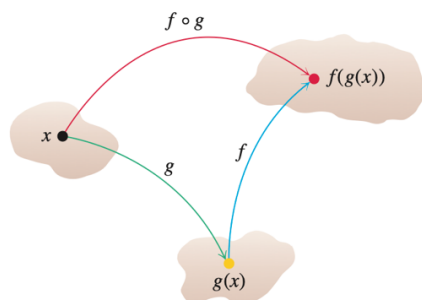


FIGURE 1.28 Arrow diagram for $f \circ g$. If x lies in the domain of g and $g(x)$ lies in the domain of f , then the functions f and g can be composed to form $(f \circ g)(x)$.

$$y = \sqrt{1-x^2}$$

$$g(x) = 1-x^2, \quad f(x) = \sqrt{x}$$

$$(f \circ g)(x) = f(g(x)) = \sqrt{1-x^2}$$

$$g(x) = 1-x^2 \quad (D_g = \mathbb{R})$$

$$f(x) = \sqrt{x} \quad (D_f = [0, \infty))$$

$$D_{fg} = [-1, 1]$$

$$1-x^2 \geq 0 \rightarrow x^2 \leq 1 \rightarrow |x| \leq 1 \rightarrow -1 \leq x \leq 1$$

Example: The function $y = \sqrt{1-x^2}$. First, we calculate $1-x^2$ and then taking the square root of the result. The function y is the composite of the function $g(x) = 1-x^2$ and function $f(x) = \sqrt{x}$.

Note that $1-x^2$ cannot be negative (range of g). The domain of the composite is $[-1, 1]$.

Example: If $f(x) = \sqrt{x}$ and $g(x) = x+1$.

- $(f \circ g)(x) = f(g(x)) = \sqrt{g(x)} = \sqrt{x+1}$. Domain: $[-1, \infty)$
- $(g \circ f)(x) = g(f(x)) = f(x) + 1 = \sqrt{x} + 1$. Domain: $[0, \infty)$
- $(f \circ f)(x) = f(f(x)) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = x^{\frac{1}{4}}$. Domain: $[0, \infty)$
- $(g \circ g)(x) = g(g(x)) = g(x) + 1 = (x+1) + 1 = x+2$. Domain: $(-\infty, \infty)$

$$f(x) = \sqrt{x}, \quad g(x) = x+1$$

$$D_g = \mathbb{R}$$

$$f \circ g(x) = f(g(x)) = \sqrt{g(x)} = \sqrt{x+1}$$

$$D_{fg} = [-1, \infty)$$

$$g \circ f(x) = g(f(x)) = f(x) + 1 = \sqrt{x} + 1$$

$$D_{gf} = [0, \infty)$$

$$(f \circ f)(x) = f(f(x)) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = x^{\frac{1}{4}}$$

$$D_{ff} = [0, \infty)$$

$$(g \circ g)(x) = g(g(x)) = g(x) + 1 = (x+1) + 1 = x+2$$

$$D_{gg} = (-\infty, \infty)$$

Shifting a Graph of a Function (pg 16)

Shifting Formulas

Vertical Shifts: $y = f(x) + k$ Shifts the graph of f **up** k units if $k > 0$
Shifts the **down** $|k|$ units if $k < 0$

Horizontal Shifts: $y = f(x+h)$ Shifts the graph of f **left** h units if $h > 0$
Shifts the **right** $|h|$ units if $h < 0$

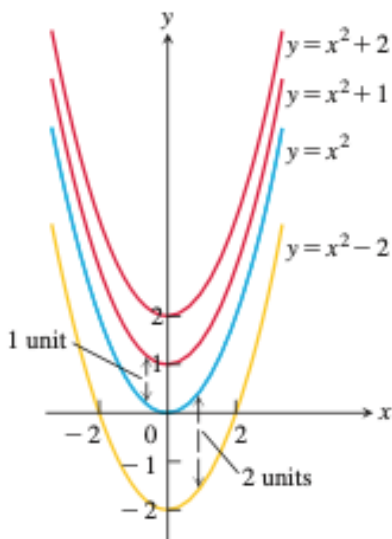


FIGURE 1.29 To shift the graph of $f(x) = x^2$ up (or down), we add positive (or negative) constants to the formula for f (Examples 3a and b).

vertical

$$y = f(x) + k$$

up k unit

$$y = f(x) + k$$

down $|k|$ unit

$$y = f(x) - k, \quad k > 0$$

horizontally

$$y = f(x)$$

$$y = f(x+h), \quad h > 0$$

left

$$x+h = x_0$$

$$x = x_0 - h$$

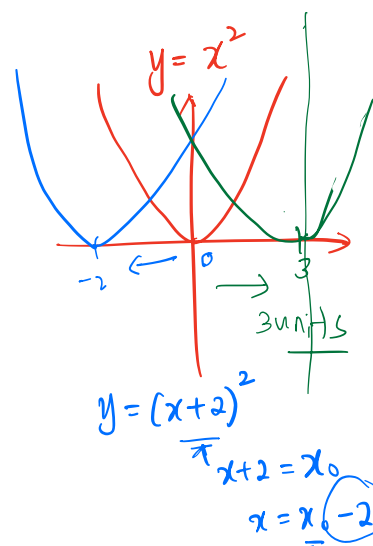
$$x+h = 0$$

$$x = -h$$

$$y = f(x-h), \quad h > 0$$

$$x-h = x_0$$

$$x = x_0 + h$$



$$y = (x-3)^2$$

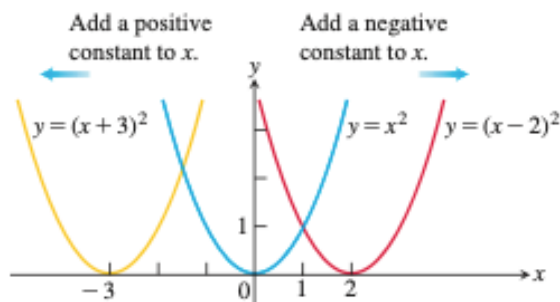


FIGURE 1.30 To shift the graph of $y = x^2$ to the left, we add a positive constant to x (Example 3c). To shift the graph to the right, we add a negative constant to x .

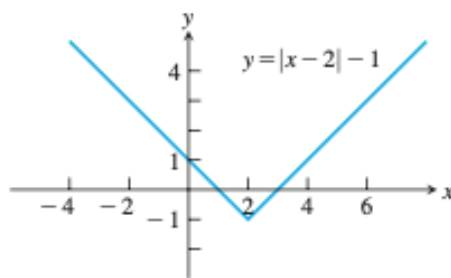
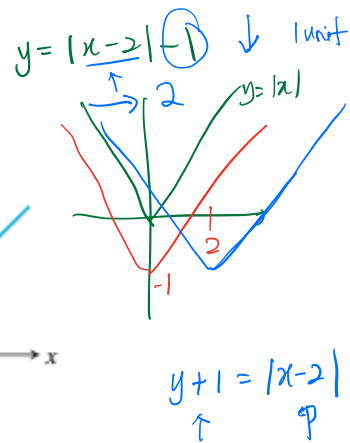


FIGURE 1.31 The graph of $y = |x|$ shifted 2 units to the right and 1 unit down (Example 3d).



Vertical and Horizontal Scaling and Reflecting Formulas

For $c > 1$, the graph is scaled:

$y = cf(x)$ Stretches the graph of f vertically by a factor of c .

$y = \frac{1}{c}f(x)$ Compresses the graph of f vertically by a factor of c .

$y = f(cx)$ Compresses the graph of f horizontally by a factor of c .

$y = f\left(\frac{x}{c}\right)$ Stretches the graph of f horizontally by a factor of c .

For $c = -1$, the graph is reflected:

$y = -f(x)$ Reflects the graph of f across the x -axis.

$y = f(-x)$ Reflects the graph of f across the y -axis.

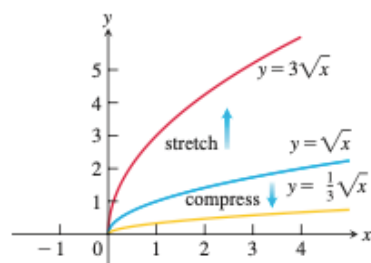
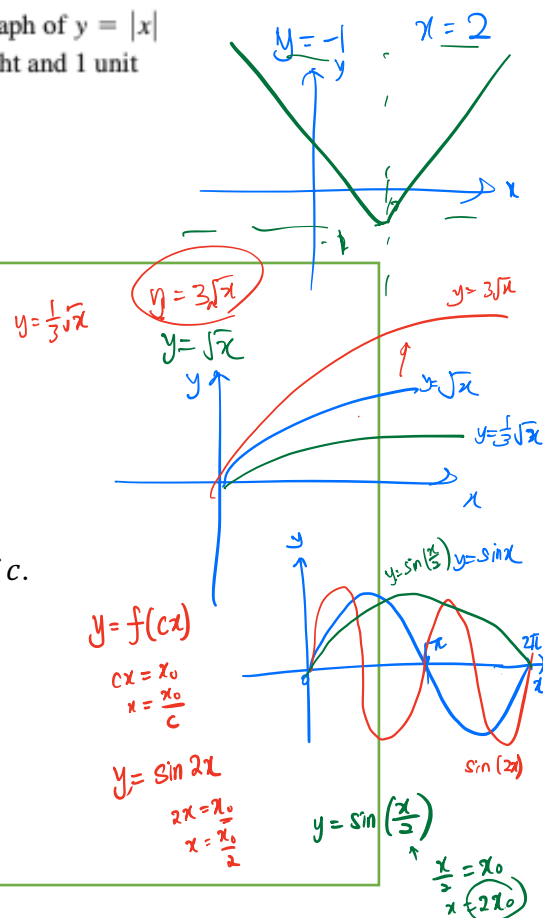


FIGURE 1.32 Vertically stretching and compressing the graph $y = \sqrt{x}$ by a factor of 3 (Example 4a).

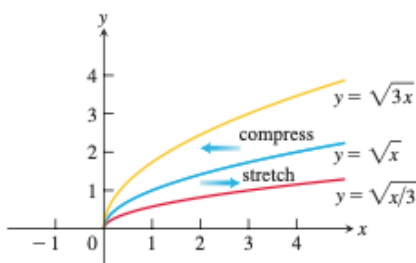


FIGURE 1.33 Horizontally stretching and compressing the graph $y = \sqrt{x}$ by a factor of 3 (Example 4b).

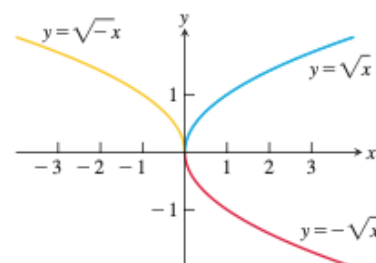


FIGURE 1.34 Reflections of the graph $y = \sqrt{x}$ across the coordinate axes (Example 4c).

$$f(x) = x^4 - 4x^3 + 10$$

$$f(-2x) = (-2x)^4 - 4(-2x)^3 + 10$$

$$= 16x^4 + 32x^3 + 10$$

EXAMPLE 5

Given the function $f(x) = x^4 - 4x^3 + 10$ (Figure 1.35a), find formulas to

- (a) compress the graph horizontally by a factor of 2 followed by a reflection across the y-axis (Figure 1.35b).
- (b) compress the graph vertically by a factor of 2 followed by a reflection across the x-axis (Figure 1.35c).

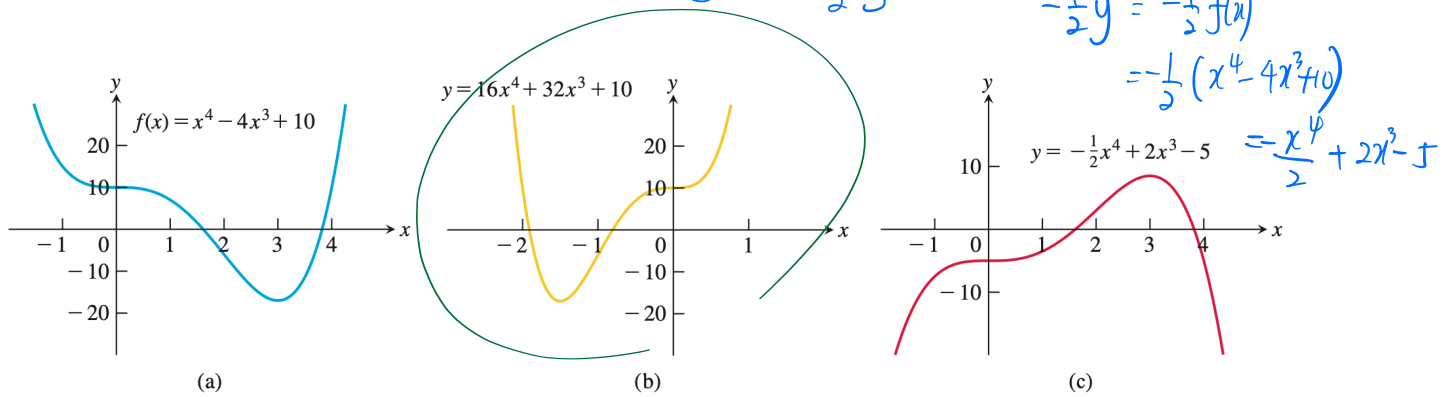


FIGURE 1.35 (a) The original graph of f . (b) The horizontal compression of $y = f(x)$ in part (a) by a factor of 2, followed by a reflection across the y-axis. (c) The vertical compression of $y = f(x)$ in part (a) by a factor of 2, followed by a reflection across the x-axis (Example 5).

Example:

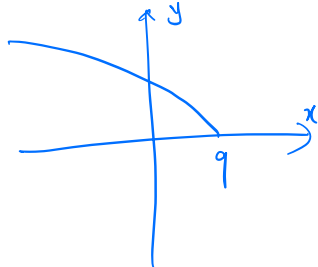
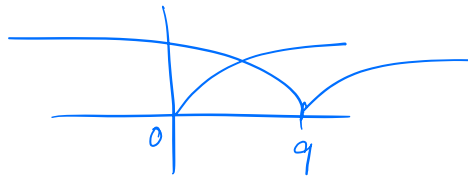


$$9 - x \geq 0 \Rightarrow x \leq 9$$

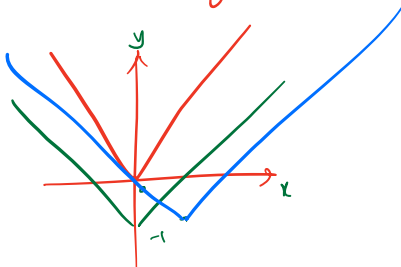
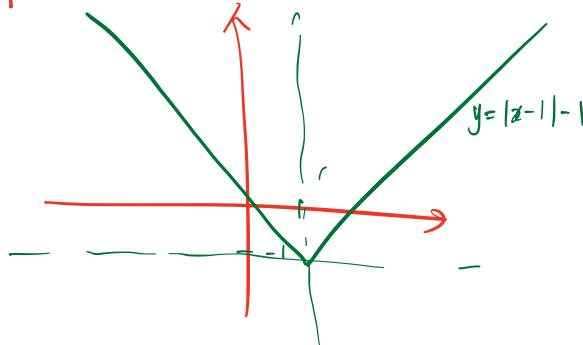
1. Graph the functions in

(a) $y = \sqrt{9 - x}$, $= \sqrt{-(x-9)}$

(b) $y = |1 - x| - 1$.



(b) $y = |1 - x| - 1 = |-(x-1)| - 1$
 $= |x-1| - 1$
 $y+1 = |x-1|$



Trigonometry Functions

Radius Measure

In calculus, it is best to see units called **radians**, because of the way they simplify later calculation.

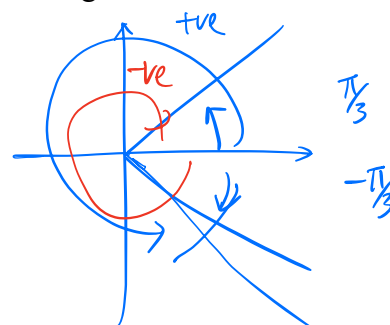
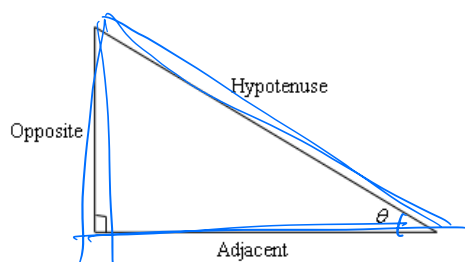
Note: π radians = 180 degrees.

$$\pi \text{ rad} = 180^\circ$$

The six basic trigonometric functions are

<u>sine</u> : $\sin \theta$	<u>cosecant</u> : $\text{cosec } \theta$ $\text{csc } \theta$
<u>cosine</u> : $\cos \theta$	<u>secant</u> : $\sec \theta$
<u>tangent</u> : $\tan \theta$	<u>cotangent</u> : $\cot \theta$

All trigonometric functions can be defined in terms of the sides of a right triangle:



$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}},$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}},$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}.$$

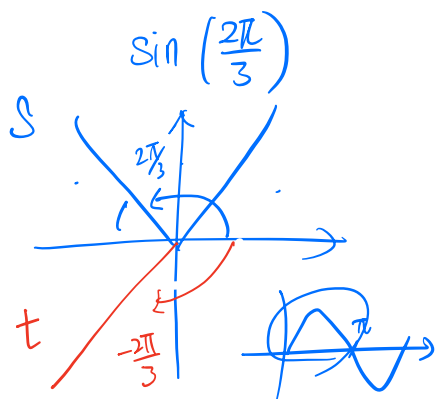
TABLE 1.1 Angles measured in degrees and radians

Degrees	-180	-135	-90	-45	0	30	45	60	90	120	135	150	180	270	360
θ (radians)	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

$$\frac{\pi}{6} = \frac{180}{6} = 30^\circ$$

Example. Evaluate $\sin\left(\frac{2\pi}{3}\right)$ and $\sin\left(-\frac{2\pi}{3}\right)$ without using the calculator.

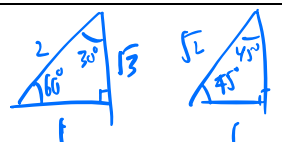
We have $\sin(2\pi/3) = \sqrt{3}/2$. The angle $-2\pi/3$ can be reached by rotating an angle of $2\pi/3$ in the clockwise direction from the positive x-axis. In doing so, we will reach the angle $4\pi/3$. Therefore, $\sin(-2\pi/3) = \sin(4\pi/3) = -\sqrt{3}/2$. Notice that $\sin(-2\pi/3) = -\sin(2\pi/3)$.



$$\sin\left(\frac{2\pi}{3}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

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$$\sin\left(-\frac{2\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

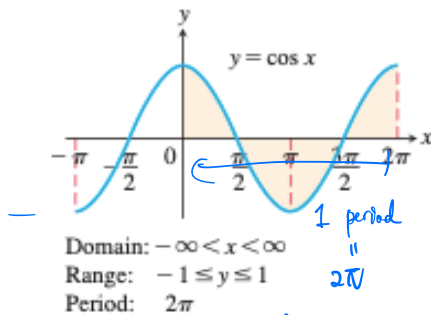


Periodicity and Graphs of the Trigonometric Functions

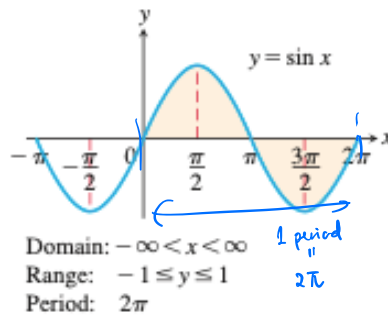
Definition: A function $f(x)$ is **periodic** if there is a positive number p such that $f(x+p) = f(x)$ or $f(x-p) = f(x)$ for every value of x . The smallest such value of p is the **period** of f .

$$\frac{1}{\cos x} = \sec x$$

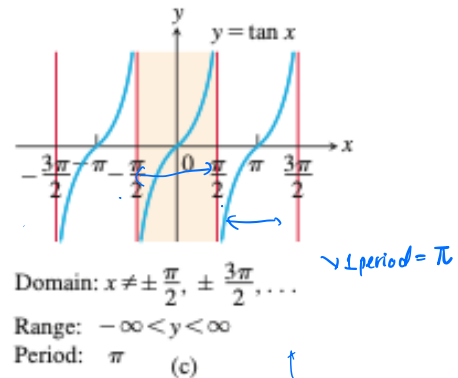
$$\frac{1}{\sin x} = \csc x$$



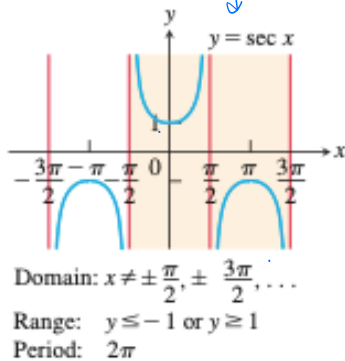
(a)



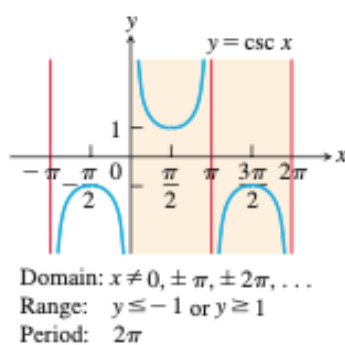
(b)



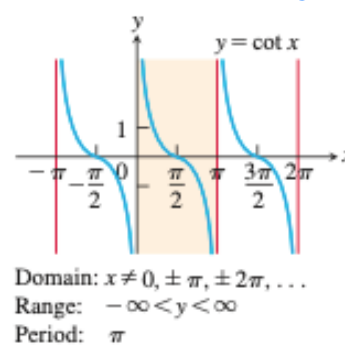
(c)



(d)



(e)



(f)

Example:

What is the period of each function?

1. $y = \sin 2x \rightarrow \pi$

2. $y = \sin\left(\frac{x}{2}\right) \rightarrow 4\pi$

3. $y = \cos\left(\frac{\pi x}{2}\right) \rightarrow 4$

4. $y = \sin\left(x + \frac{\pi}{6}\right) \rightarrow 2\pi$

Even:

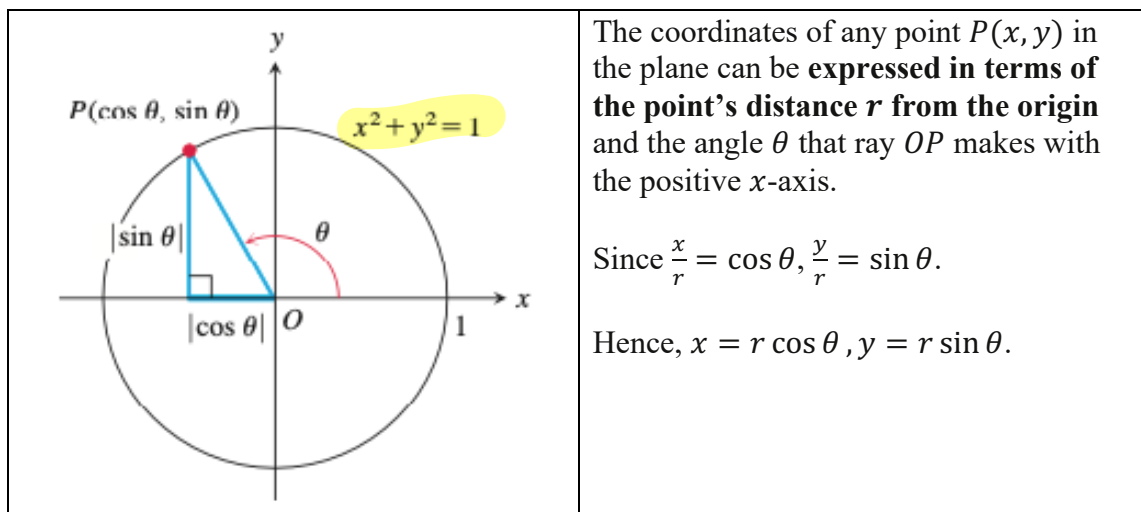
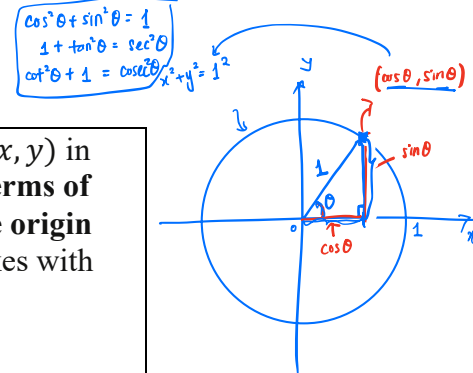
$$\rightarrow \cos(-x) = \cos x$$

Odd:

$$\sin(-x) = -\sin x$$

$$\tan(-x) = -\tan x$$

$$\begin{aligned} \tan x &= \frac{\sin x}{\cos x} \\ \tan(-x) &= \frac{\sin(-x)}{\cos(-x)} \rightarrow \text{odd} \\ &= \frac{-\sin x}{\cos x} = -\tan x. \end{aligned}$$



Trigonometric Identities

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

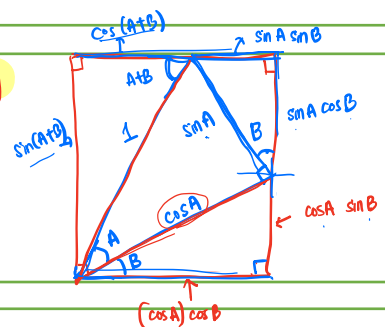
$$1 + \cot^2 \theta = \csc^2 \theta$$

Addition Formulas

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$



Double-Angle Formulas

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\begin{aligned} \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ \cos 2\theta &= 1 - 2 \sin^2 \theta \\ \cos 2\theta &= 2 \cos^2 \theta - 1 \end{aligned}$$

Half-Angle Formulas

$$\cos^2(\theta) = \frac{1 + \cos 2\theta}{2}$$

$$\sin^2(\theta) = \frac{1 - \cos 2\theta}{2}$$

The Law of Cosines

$$c^2 = a^2 + b^2 - 2ab \cos \theta$$

$$\begin{aligned}\sin(\pi - \theta) &= \sin \theta \\ \cos(\pi - \theta) &= -\cos \theta\end{aligned}$$

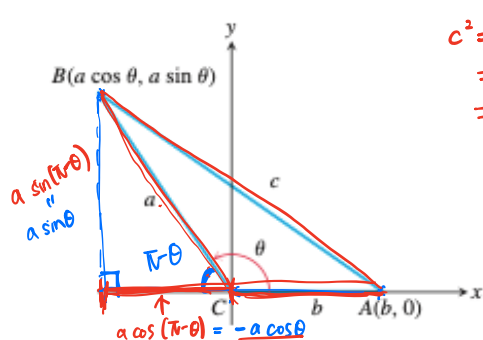


FIGURE 1.46 The square of the distance between A and B gives the law of cosines.

$$\begin{aligned}c^2 &= (a \sin \theta)^2 + (b - a \cos \theta)^2 \\ &= a^2 \sin^2 \theta + b^2 - 2ab \cos \theta + a^2 \cos^2 \theta \\ &= a^2 + b^2 - 2ab \cos \theta.\end{aligned}$$

Two Special Inequalities

$$\rightarrow -|\theta| \leq \sin \theta \leq |\theta| \text{ and } -|\theta| \leq 1 - \cos \theta \leq |\theta|$$

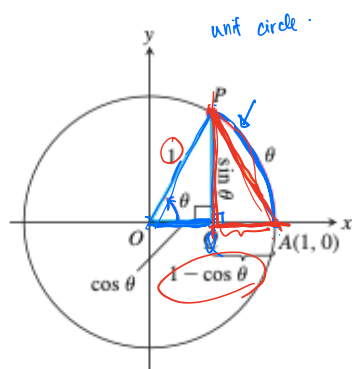


FIGURE 1.47 From the geometry of this figure, drawn for $\theta > 0$, we get the inequality $\sin^2 \theta + (1 - \cos \theta)^2 \leq \theta^2$.

$$\begin{aligned}|\Delta| &= \theta \\ |AP| &\leq \theta \\ |AP|^2 &= \sin^2 \theta + (1 - \cos \theta)^2 \leq \theta^2 \\ \sqrt{\sin^2 \theta + (1 - \cos \theta)^2} &\leq \sqrt{\theta^2} \\ |\sin \theta| &\leq |\theta| \\ \Rightarrow -|\theta| &\leq \sin \theta \leq |\theta|\end{aligned}$$

Examples (exercise 1.3):

1. Solve $\cos 2\theta + \cos \theta = 0$ for $0 \leq \theta \leq 2\pi$.

2. Evaluate $\cos\left(\frac{\pi}{12}\right)$.

① $\cos 2\theta + \cos \theta = 0$

$$2\cos^2 \theta - 1 + \cos \theta = 0$$

$$2\cos^2 \theta + \cos \theta - 1 = 0$$

$$y = \cos \theta$$

$$2y^2 + y - 1 = 0$$

$$(2y - 1)(y + 1) = 0$$

$$y = \frac{1}{2} \text{ or } y = -1$$

$$\cos \theta = \frac{1}{2}$$

$$\cos \theta = -1$$

$$0 \leq \theta \leq 2\pi$$



$$\cos \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\theta = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$$

$$\cos\left(\frac{\pi}{12}\right)$$

$$= \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$$

$$= \cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4}$$

$$= \frac{1}{2} \left(\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$